

Duplicate leakage in a public conjunctival pallor benchmark, and an honest baseline for image-based anaemia screening

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August 31, 2026

Abstract

Conjunctival pallor photographs are an attractive substrate for non-invasive anaemia screening, and CP-AnemiC — 710 images of Ghanaian children aged 6–59 months with paired laboratory haemoglobin — has become a convenient public benchmark for the task. We report that the benchmark contains duplicate records substantial enough to change published conclusions, and we provide a corrected baseline.

Of the 710 metadata rows only 479 are distinct photographs: 212 files are byte-identical copies registered under different image identifiers, and a further 19 are pixel-identical after region-of-interest extraction while differing at the byte level, so content hashing alone does not detect them. Ninety duplicate groups carry *conflicting* haemoglobin labels on identical pixels, one spanning 3.3 to 10.4 g/dL. Evaluating with a random split therefore trains and tests on the same photograph and inflates AUROC by +0.103 (95 % CI +0.056 to +0.152; bootstrap $p < 0.001$, $n_{\text{boot}} = 5000$).

After removing this redundancy and grouping cross-validation by collection site, twelve interpretable colour features reach AUROC 0.780 (95 % CI 0.737–0.818) for WHO anaemia (Hb < 11 g/dL), against a demographics-only floor of 0.565; the image contributes +0.215 AUROC (DeLong 95 % CI +0.151 to +0.279, $p = 4.4 \times 10^{-11}$). Bland–Altman limits of agreement nonetheless span 7.3 g/dL with strong proportional bias (slope -0.67), so the model is a triage screen and not a haemoglobin meter. At a threshold catching 90 % of anaemic children, specificity falls to 0.362, and decision curve analysis shows the model is worse than referring every child below a threshold probability of about 0.15 — so it earns its place only where confirmatory testing is scarce.

We recommend that future work on CP-AnemiC deduplicate at the pixel level and group cross-validation by site, and we release the full analysis so that reported numbers can be checked against these controls.

Keywords: anaemia screening; conjunctival pallor; data leakage; benchmark integrity; medical imaging; low-resource diagnostics.

1 Introduction

Anaemia affects an estimated 40 % of children aged 6–59 months worldwide, rising to 68 % in west and central Africa [2], and diagnosis conventionally requires a blood draw and laboratory or point-of-care haemoglobin assay. Both are constrained in low-resource settings by cost, consumables,

trained staff and infection-control requirements. This has motivated interest in image-based screening: the palpebral conjunctiva is a mucous membrane whose colour is visibly influenced by haemoglobin concentration, is straightforward to expose, and can be photographed with a consumer smartphone.

Public datasets are what allow such methods to be compared. CP-AnemiC [1] pairs conjunctiva photographs of Ghanaian children with laboratory haemoglobin and has been adopted as a benchmark for this task. The value of any benchmark, however, rests on an assumption that is rarely re-examined once a dataset is in circulation: that its records are distinct. Where they are not, a randomly partitioned evaluation places copies of the same photograph in both training and test folds, and the resulting score measures memorisation rather than generalisation.

This paper makes three contributions.

1. We show that CP-AnemiC contains duplicate records at a scale that materially inflates reported performance, and we quantify that inflation.
2. We establish a corrected baseline under pixel-level deduplication and site-grouped cross-validation, using interpretable colour features only.
3. We characterise the resulting model as a screening instrument — reporting the operating point, calibration and Bland–Altman agreement that determine whether it could be deployed — and conclude that it is not yet adequate as a haemoglobin estimator.

Our aim is deliberately not a new architecture. It is to establish what the honest number on this benchmark is, and to make the controls that produce it reproducible.

2 Data

2.1 Source

CP-AnemiC [1] comprises conjunctiva photographs of children aged 6–59 months collected at ten hospitals across four Ghanaian regions, each paired with a laboratory haemoglobin measurement. We use the WHO threshold for childhood anaemia, $\text{Hb} < 11 \text{ g/dL}$ [3]. Table 1 summarises the dataset as distributed and after the deduplication described below.

Table 1: CP-AnemiC as distributed and after deduplication.

Property	Value
Metadata rows (as distributed)	710
Distinct images by SHA-256	498
Distinct images at pixel level	479
Rows appearing exactly once	407
Duplicate groups with conflicting Hb	90
Collection sites	10 hospitals, 4 regions
Cohort	children 6–59 months
Label	laboratory Hb (g/dL); WHO threshold 11
Anaemia prevalence after dedup	0.493

2.2 Duplicate structure

Two distinct forms of redundancy are present. *Byte-identical* duplication accounts for 212 files: the same image registered under more than one identifier, detectable by hashing the file. A further 19 images are *pixel-identical after region-of-interest extraction* but differ at the byte level — re-encoded or re-saved copies — and are therefore invisible to content hashing. Detecting

these requires comparing decoded pixels within the annotated region, which is how the count falls from 498 to 479.

The consequential subset is the 90 duplicate groups carrying conflicting haemoglobin labels on identical pixels. One such group spans 3.3 to 10.4 g/dL. Because the pixels are the same, no model can satisfy both labels; the disagreement is irreducible label noise that places a ceiling on achievable performance. Where duplicates were merged we took the median haemoglobin, which is a defensible summary but cannot recover the truth.

2.3 Mask handling

Each image is RGBA, where the alpha channel is a hand-drawn conjunctiva mask covering approximately 25 % of the frame; the remainder is black padding. Colour statistics must therefore be computed over masked pixels only. Ignoring the mask shifts mean RGB by more than 100/255 and produces features that encode *mask area* — an artefact of annotator behaviour — rather than tissue colour.

3 Methods

3.1 Features

We extract twelve interpretable colour features over the masked region of interest: mean R , G , B ; circular mean hue and hue concentration; mean saturation; CIELAB L^* , a^* , b^* ; an illumination-normalised redness index $r/(r + g + b)$; its 10th percentile; and the red/green ratio. The a^* channel is the perceptual red–green axis that corresponds to what a clinician reads as pallor.

Two exclusions are deliberate. **Mask area is not used as a feature**: it reflects how the annotator drew the boundary rather than physiology, and had it correlated with site or severity the model would have learned the annotator. **Hue is averaged circularly**: red tissue sits near both 0.0 and 1.0 on the hue circle, so a linear mean of 0.98 and 0.02 returns 0.5, which is cyan.

3.2 Model and validation

A gradient-boosted regressor predicts haemoglobin; negating its output yields a screening score. Hyperparameters were fixed *a priori* and never tuned on this data; a nested cross-validation check (Section 4.10) confirms this cost essentially nothing. All reported metrics are out-of-fold.

The validation protocol combines two requirements. Deduplication must operate at the pixel level, for the reasons above. Cross-validation folds must be grouped by collection site, so that a model cannot exploit site-specific camera, lighting or operator signatures shared between training and test folds. We report a waterfall over both.

4 Results

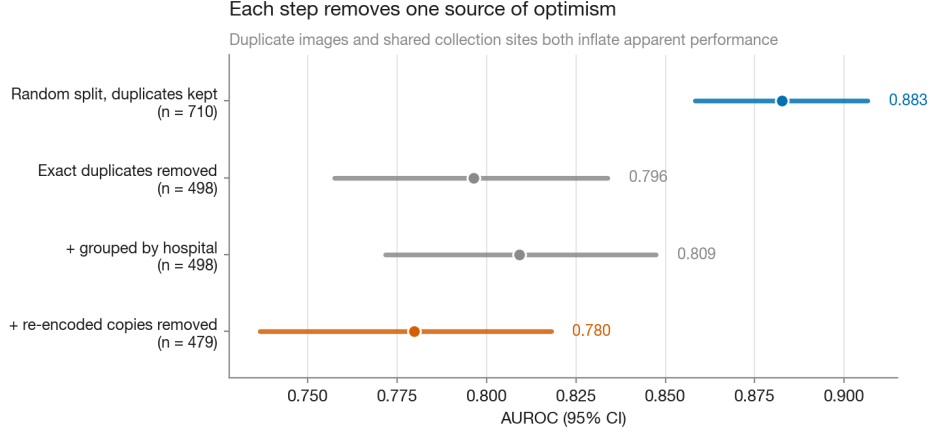
4.1 Progressive removal of optimism

Table 2 evaluates the same colour features under successively stricter protocols. Each row closes one more leak than the row above.

The naive configuration — a random split with duplicates retained, which is the default when a dataset is assumed clean — reports AUROC 0.883. The corrected protocol reports 0.780. A paired bootstrap over 5000 resamples estimates the inflation at +0.103 (95 % CI +0.056 to +0.152, $p < 0.001$). Figure 1 shows the waterfall.

Table 2: Effect of evaluation protocol on apparent performance. Colour features only.

Configuration	n	AUROC [95 % CI]	MAE (g/dL)
Random split, duplicates kept (<i>common setup</i>)	710	0.883 [0.858–0.906]	1.46
Exact duplicates removed, random split	498	0.796 [0.758–0.834]	1.44
+ grouped by collection site	498	0.809 [0.772–0.847]	1.42
+ pixel-level duplicates removed (headline)	479	0.780 [0.737–0.818]	1.46

**Figure 1:** Apparent performance falls as each source of leakage is removed. The gap between the first and last bars is the optimism introduced by evaluating on a benchmark whose records are not distinct.**Table 3:** Feature ablation under the corrected protocol ($n = 479$). Comparisons by DeLong’s test [5].

Feature set	AUROC [95 % CI]	vs. colour
Demographics only (age, sex)	0.565 [0.514–0.618]	$\Delta - 0.215, p = 4.4 \times 10^{-11}$
Colour only	0.780 [0.737–0.818]	—
Colour + demographics	0.769 [0.725–0.808]	$\Delta - 0.011, p = 0.30$

4.2 Does the image contribute?

A screening score could in principle be driven by demographics alone. Table 3 rules this out.

The image carries substantial and highly significant signal over demographics. Adding demographics to colour produces no detectable improvement, which is the expected result if the photograph already encodes what age and sex would otherwise proxy. Figure 2 shows the corresponding ROC curves.

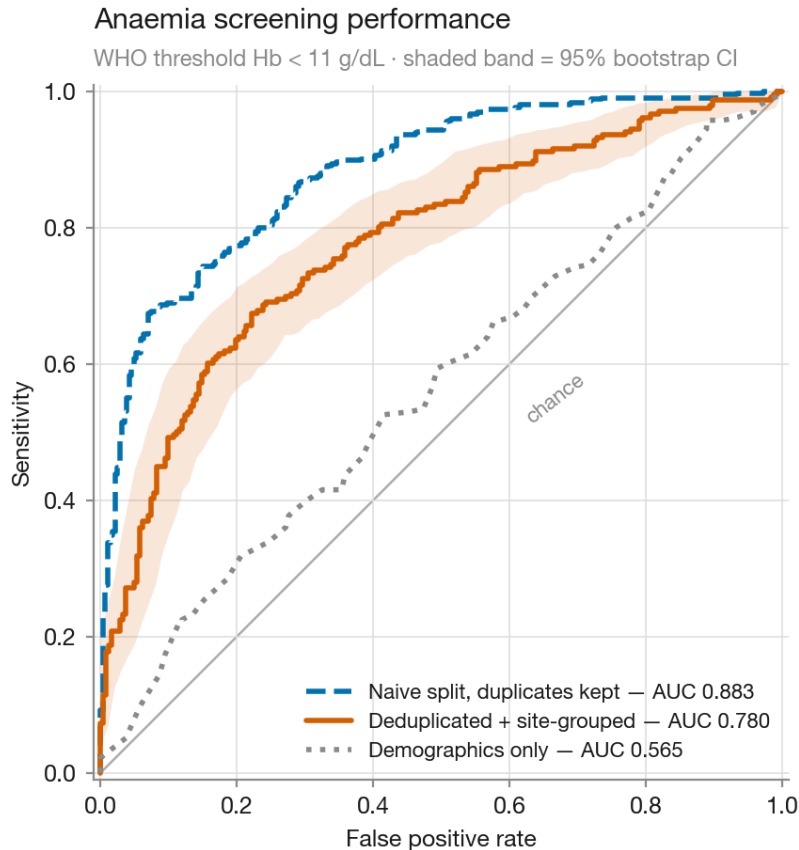


Figure 2: ROC curves under the corrected protocol. Colour features clearly separate from the demographics-only floor; colour + demographics is indistinguishable from colour alone.

4.3 Choice of model

We report both a haemoglobin regressor and a direct binary classifier, because they answer different questions and we did not wish to adopt the better number after the fact.

Table 4: Regressor versus direct classifier.

Model	AUROC [95 % CI]	Hb output	Agreement assessable
Hb regressor (<i>headline</i>)	0.780 [0.737–0.818]	yes, MAE 1.46 g/dL	yes
Binary classifier	0.818 [0.778–0.854]	no	no

The classifier discriminates better, as expected when the decision is optimised directly, but it produces no haemoglobin estimate and therefore cannot be examined for clinical agreement. We retain the regressor as the headline because the agreement result (Section 4.6) is the more consequential finding.

4.4 Calibration

Table 5: Post-hoc calibration degrades both calibration and discrimination. Isotonic calibration applied via `CalibratedClassifierCV` with $cv = 3$.

Variant	Brier	ECE	H-L p	AUROC	prob. range
Uncalibrated	0.174	0.054	0.021	0.818	[0.01, 0.98]
Isotonic	0.200	0.116	< 0.001	0.788	[0.00, 0.92]

Wrapping the model in post-hoc calibration made both calibration and discrimination worse. The mechanism is visible in the probability range: `CalibratedClassifierCV` fits k sub-models on $k - 1$ folds each and averages them, pulling probabilities toward the centre, and at roughly 380 training rows each sub-model is additionally data-starved. The result is systematic *under*-confidence — the opposite of the over-confidence that calibration is normally applied to cure. We therefore report the uncalibrated model, which remains slightly imperfect (Hosmer–Lemeshow $p = 0.021$). We note this because the standard recommendation to calibrate is not unconditional at this sample size.

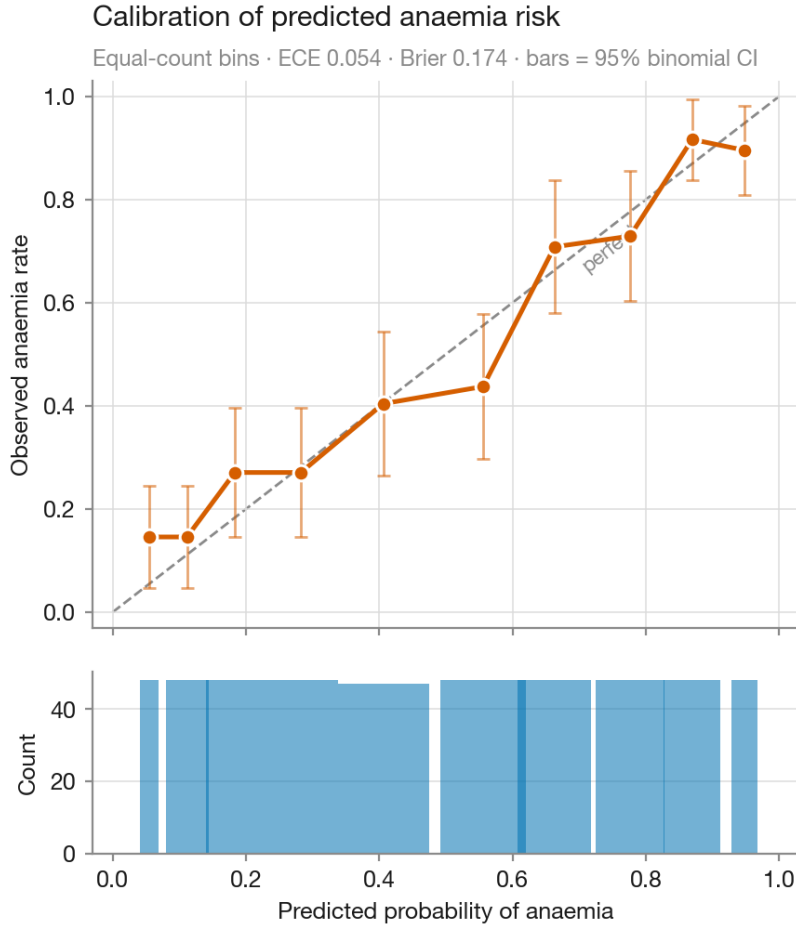


Figure 3: Reliability curves. Post-hoc isotonic calibration compresses the probability range and worsens both Brier score and expected calibration error.

4.5 Screening operating point

At the WHO threshold the model achieves sensitivity 0.775 and specificity 0.630. Tuned to catch 90% of anaemic children, specificity falls to 0.362 — referring roughly 64% of healthy

children for confirmatory testing. This, rather than AUROC, is the quantity that determines deployability, and at present it is not good enough for unsupervised use.

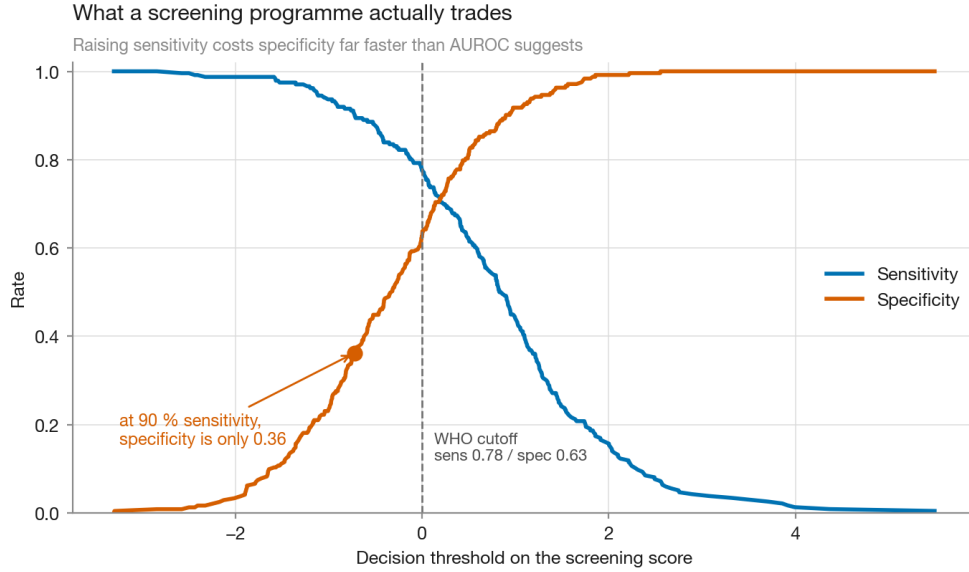


Figure 4: Sensitivity and specificity against the decision threshold, from the same out-of-fold predictions as the headline result. The curves cross steeply: moving from the WHO cutoff to 90 % sensitivity costs roughly half the specificity. This trade, rather than AUROC, is what a screening programme has to absorb.

4.6 Agreement as a haemoglobin estimator

Bland–Altman analysis [6] gives a bias of +0.03 g/dL with limits of agreement from −3.65 to +3.70 g/dL, a span of 7.3 g/dL, and a proportional-bias slope of −0.67. The model shrinks toward the population mean, over-predicting in anaemic children and under-predicting in healthy ones. A clinically useful non-invasive haemoglobin meter would require limits nearer ± 1 g/dL. **No haemoglobin value from this model should be reported to a clinician.**

4.7 Is acting on the model better than the alternatives?

Discrimination and calibration describe the model; neither says whether acting on it beats what a clinic would otherwise do. Where the realistic comparators are “refer every child” and “refer none”, that comparison is what decides deployment. Decision curve analysis [4] makes it directly, through net benefit

$$\text{NB}(p_t) = \frac{\text{TP}}{N} - \frac{\text{FP}}{N} \cdot \frac{p_t}{1 - p_t}, \quad (1)$$

where p_t is the probability at which referral becomes worthwhile and the odds term prices one missed case against the false referrals a clinic will tolerate for it. A model is worth using only where its curve lies above both comparators.

Figure 7 shows the result, and it is a qualified one. **Below $p_t \approx 0.15$ the model is worse than simply referring every child.** That is a direct consequence of the 49 % prevalence in this cohort: when half the children are anaemic and missing a case is judged far costlier than an unnecessary referral, no triage rule beats referring everyone. Above that threshold the model does add benefit, and the margin widens as false referrals become more costly — at $p_t = 0.4$ its net benefit is 0.29 against 0.15 for referring everyone.



Figure 5: Predicted against laboratory haemoglobin. A least-squares fit through the predictions has slope 0.28, far below the identity line: a 1 g/dL change in true haemoglobin moves the prediction by roughly a quarter of that. The model separates anaemic from non-anaemic children while carrying little information about *how* anaemic they are.

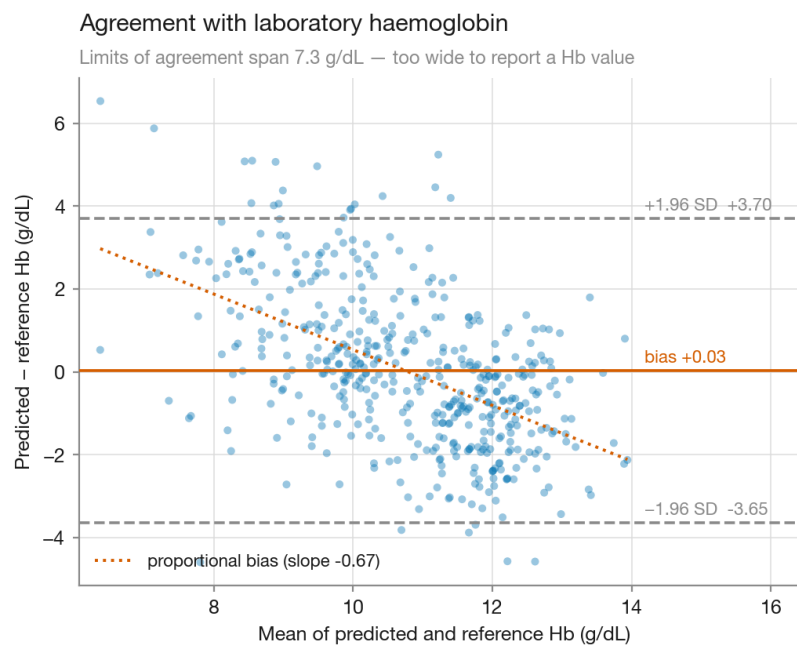


Figure 6: Bland–Altman agreement. The limits span 7.3 g/dL and the proportional bias is pronounced, establishing this as a triage screen rather than a measurement device.

The practical reading is that this model earns its place only where confirmatory testing is scarce enough that indiscriminate referral is not an option. That is a narrower claim than AUROC 0.780 suggests on its own, and it is the claim the evidence supports.

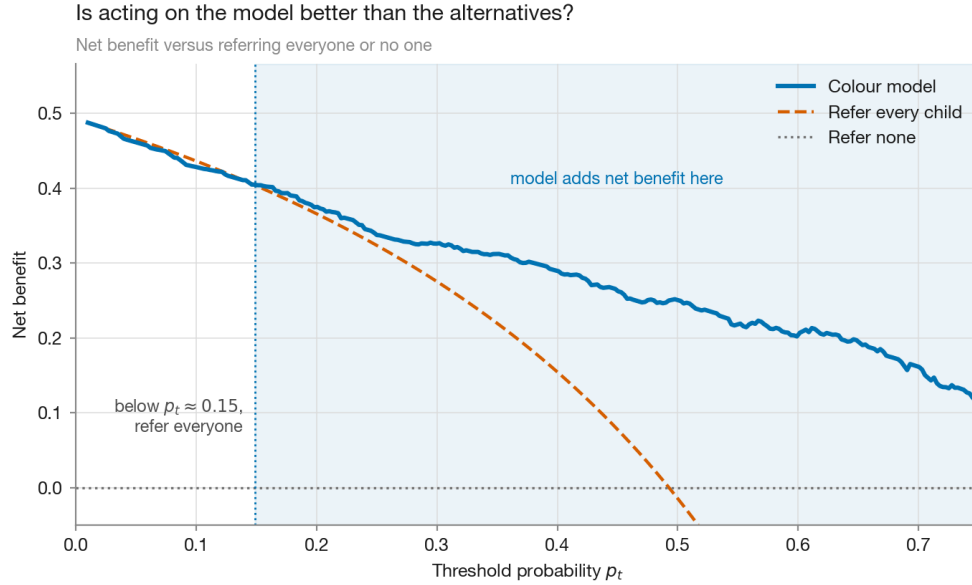


Figure 7: Decision curve analysis. Net benefit of the colour model against the two comparators available to a clinic. The model is not universally useful: below $p_t \approx 0.15$, referring every child is the better strategy, a consequence of the high prevalence in this cohort. Its advantage appears once false referrals carry real cost.

4.8 Cross-site generalisation

Leave-one-site-out evaluation across nine evaluable hospitals gives a mean AUROC of 0.820 (range 0.688–1.000). The perfect score at SDA Hospital arises from $n = 11$ at 82 % prevalence and should be read as noise rather than performance; the informative observation is that the *lowest* score belongs to the *largest* site (Komfo Anokye Teaching Hospital, $n = 94$, AUROC 0.703), which is the direction one would expect if smaller sites are optimistically estimated.

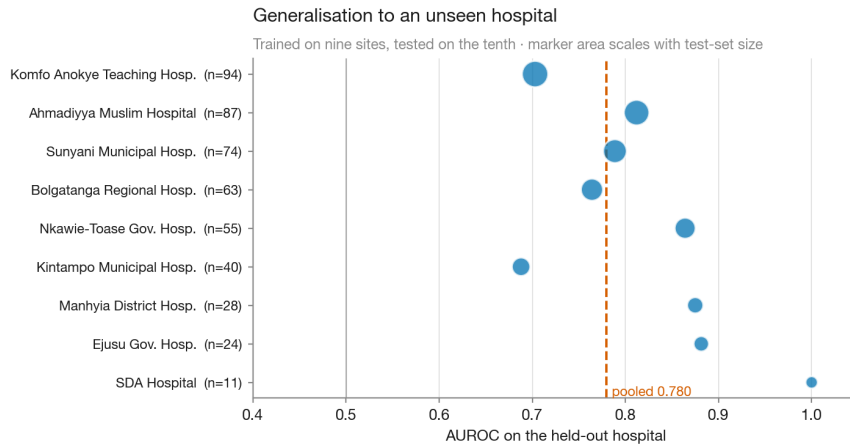


Figure 8: Leave-one-site-out performance. Marker size reflects site n ; small sites carry wide, unreliable intervals.

4.9 Which features carry the signal

Permutation importance was measured out-of-fold, because the model’s internal impurity importances are computed on training data and are biased toward continuous features. Saturation (Δ AUROC 0.073) and the 10th percentile of the redness index (0.055) dominate, followed by circular mean hue (0.040), the redness index (0.039) and mean red (0.034). Saturation and the pale tail of the redness distribution are the physiologically expected carriers of pallor. Because these features are strongly correlated, a low score should be read as “redundant given the others”, not “irrelevant”. Figure 9 shows this rather than assuming it: mean absolute pairwise correlation across the twelve features is 0.51, reaching 0.98 between mean green and CIELAB L^* . Two clusters are visible — an RGB/lightness group and a saturation-and-redness group — and permutation importance divides credit within each. That is why no single feature scores highly even though the set as a whole carries +0.215 AUROC over demographics.

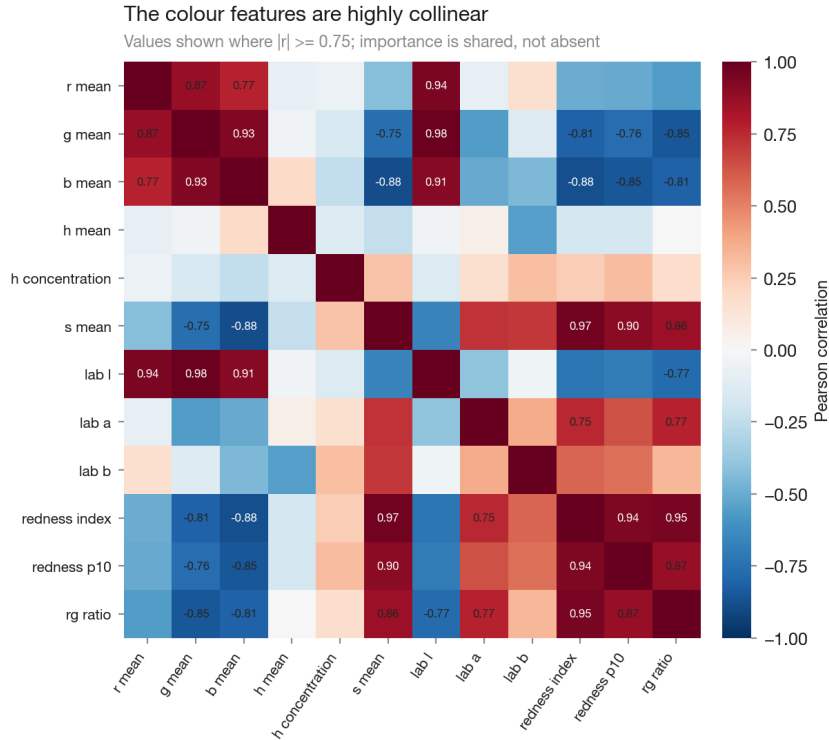


Figure 9: Pairwise correlation between the twelve colour features, printed where $|r| \geq 0.75$. This collinearity is why a low permutation-importance score should be read as redundancy given the rest of the set rather than absence of signal.

4.10 Controls

Table 6 lists checks designed to make the headline result fail if it were an artefact.

Table 6: Adversarial and robustness controls.

Control	Result	Interpretation
Label permutation (10 repeats)	AUROC 0.507 ± 0.026	chance; the harness does not leak
Seed stability (10 seeds)	0.778 ± 0.003	not fold-assignment luck
Bootstrap vs. DeLong CI	$[0.737, 0.818]$ vs. $[0.738, 0.821]$	independent machinery agrees
Unambiguous singles only ($n = 407$)	0.763	holds where no label conflict exists
Nested-CV hyperparameter tuning	$0.780 \rightarrow 0.782$	fixed settings not cherry-picked
Duplicate-threshold sweep (0.5–5.0)	0.774–0.784	conclusion not sensitive to cutoff

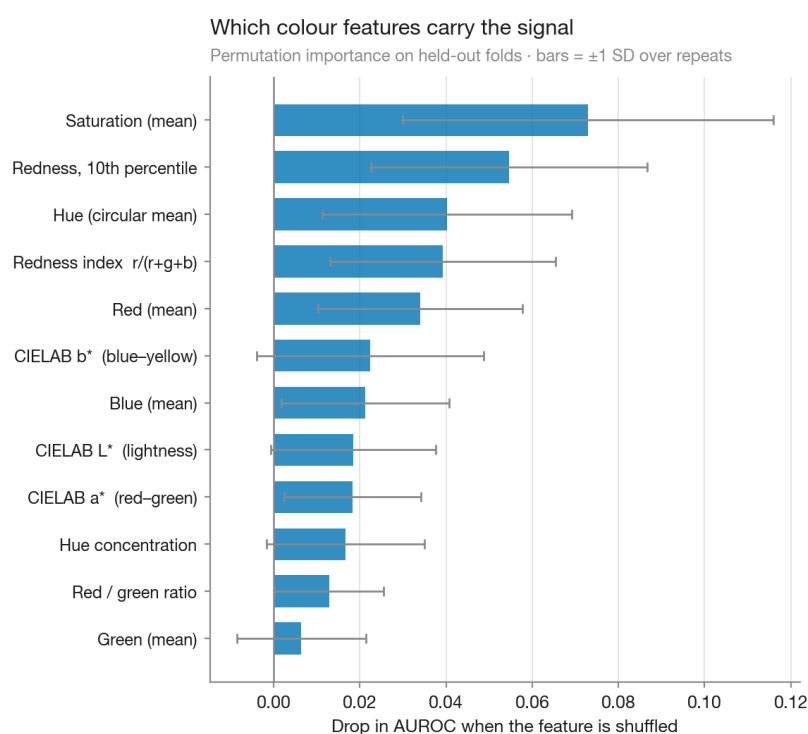


Figure 10: Out-of-fold permutation importance. Saturation and the pale tail of the redness distribution dominate. Correlated features share credit, so a low score means redundant given the others rather than uninformative.

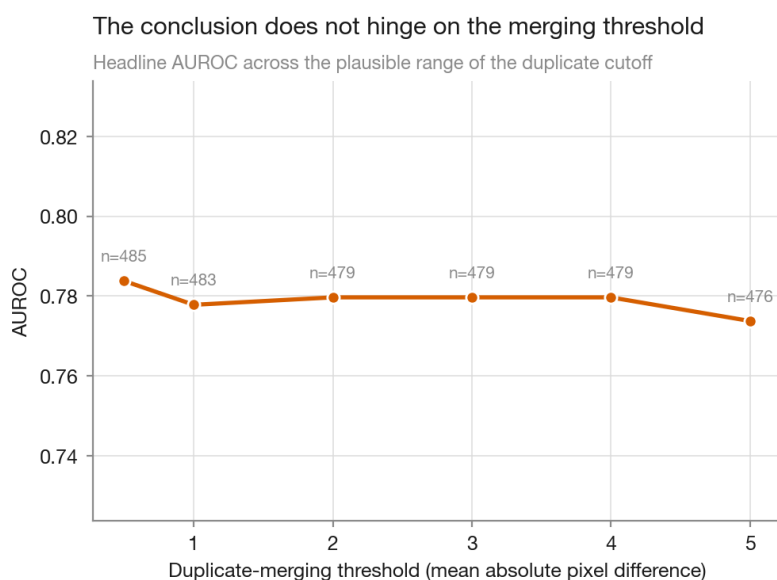


Figure 11: Sensitivity of the headline result to the perceptual duplicate-detection threshold. AUROC varies between 0.774 and 0.784 across the sweep, so the conclusion does not depend on where the cutoff is placed.

4.11 Would more data help?

A learning curve over training-set size rises from AUROC 0.635 at 120 images to 0.782 at 479 and is still climbing at full size. Performance here is therefore limited by sample size as well as by the dataset’s irreducible label noise; a larger, cleanly labelled dataset would plausibly exceed 0.780.

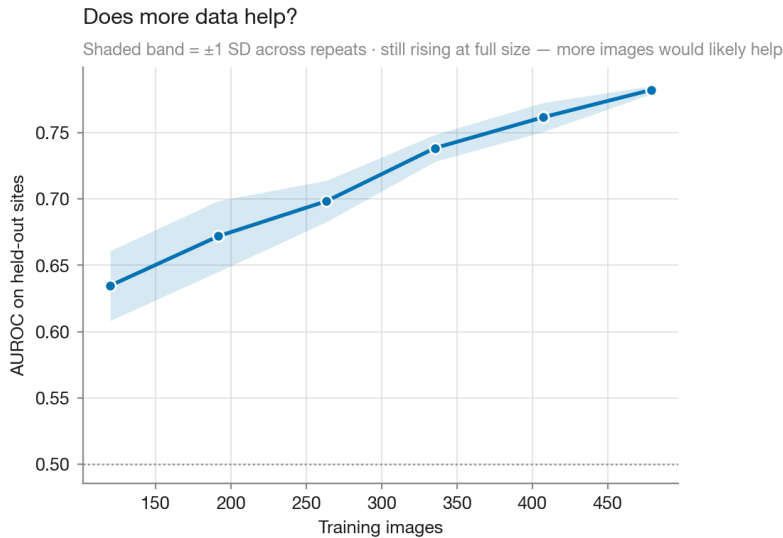


Figure 12: Learning curve. The absence of a plateau indicates the benchmark is sample-limited.

5 Discussion

The gap between 0.883 and 0.780 is not a modelling result; it is an artefact of dataset handling. It matters because it is the kind of gap that separates a method reported as promising from one reported as marginal, and because nothing about the naive protocol looks careless from the inside — a random split is the default, and duplicates are invisible unless specifically sought. The 19 images that are pixel-identical only after region-of-interest extraction are a particularly awkward case: a reasonable practitioner who deduplicates by file hash would still be leaking.

Our corrected baseline should be read as establishing a floor for interpretable colour features, not a ceiling for the task. Learned representations may well exceed it. The purpose of reporting it is that any such claim can now be measured against a number produced under controls that are stated and reproducible.

The clinical framing deserves emphasis. Discrimination at AUROC 0.780 is not uninteresting for triage in a setting where the alternative is no screening at all. But the Bland–Altman result forecloses use as a haemoglobin estimator, and the operating point — 64 % of healthy children referred at 90 % sensitivity — would impose a confirmatory testing burden that may exceed what a low-resource clinic can absorb. Both numbers are more decision-relevant than the AUROC that such studies typically headline.

6 Limitations

- **Label noise is irreducible.** Ninety duplicate groups carry conflicting haemoglobin on identical pixels; median merging cannot recover the truth, and a performance ceiling is therefore built into the benchmark.

- **No colour calibration.** No colour card or white-balance reference was captured, so absolute RGB partly encodes illumination. Normalised ratios mitigate but do not eliminate this.
- **One country, one age band.** Children 6–59 months in Ghana. Nothing here supports use in adults, in pregnancy, or across different skin tones and camera pipelines.
- **Hand-drawn masks.** A deployed system would require automatic conjunctiva segmentation, whose error is not represented in any number reported here.
- **No true external validation.** Leave-one-site-out varies camera, operator and prevalence, but not country or protocol. Validation on an independent dataset is the obvious next step.
- **Not a clinical device.** No regulatory, ethical or safety assessment has been performed. This is a software study on a public dataset.

7 Conclusion

CP-AnemiC contains duplicate records — including duplicates that survive content hashing and duplicates carrying conflicting labels — at a scale that inflates AUROC by approximately 0.10 under a random split. We recommend that future work on this benchmark deduplicate at the pixel level within the annotated region and group cross-validation folds by collection site.

Under those controls, twelve interpretable colour features achieve AUROC 0.780 (95 % CI 0.737–0.818) for WHO childhood anaemia, with the image contributing +0.215 over demographics alone. The resulting model is a plausible triage screen and is not a haemoglobin meter: limits of agreement span 7.3 g/dL and specificity falls to 0.362 at 90 % sensitivity. Reporting these quantities alongside AUROC is, we suggest, the minimum needed to judge whether such a method could be deployed.

Use of generative AI

Generative AI (Anthropic Claude) was used to draft and typeset this manuscript from the author’s own analysis, results and written findings, and to assist with the accompanying analysis code. The study design, the dataset audit and duplicate-leakage discovery, the choice of controls, and the interpretation of results are the author’s own. The author has reviewed and verified the entire manuscript, including every reported figure, and takes full responsibility for its content.

Reproducibility

The complete analysis runs in approximately 80 seconds on a laptop:

```
pip install -r requirements.txt
# download CP-AnemiC into data/cp-anemic/
python3 scripts/run_full_analysis.py
python3 -m pytest tests/ -q
```

Every number and figure in this paper is regenerated by that script. The accompanying test suite (62 tests) is organised by failure mode — split integrity, feature correctness, numerical robustness, determinism, statistical machinery, and adversarial nulls — on the principle that the dangerous failure in this setting is not a crash but a plausible wrong number.

Data availability

CP-AnemiC is publicly available from Mendeley Data ([m53vz6b7fx](#)) [1] and is used under its published licence. No patient-identifiable data is redistributed. Derived colour statistics only are stored in the accompanying repository.

Conflicts of interest

None declared.

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